STUDY OF DIFFERENT TYPES OF CABLE NETWORK

Exp no: 2

Date:

Aim:

To study various network cable types, understanding their specifications, and construct a functional straight-through or crossover Ethernet cable using RJ45 connectors

Tools required:

* Cable: CAT5 or CAT5e Ethernet cabling
* Connectors: Two RJ45 plus Cp
* Tools: Modular crimping tool

PART-A. Network Cable Comparison.

Cable Type	Standard/ category	Max Data Transmission	Advantages/ Disadvantage	Application
UTP	Category 3 category 5 category 5e	10Mbps 100Mbps 1Gbps	Adv: Low cost, high flexibility, easy installation disadv: High susceptibility	10Base-T, Fast Ethernet
STP/SSTP	Category 6 Category 7	10 Gbps 10Gbps	Adv: Shielded against, higher speed Disadv: Higher cost	Data centers, 10G Ethernet
Coaxial Cable	RG-6, RG-59, RG-11	10-100Mbps	Adv: High bandwidth, Immune to basic EMI, versatile. Disadv: Bulky size, higher relative cost	Cable TV networks, high-speed broadband connections
Fibre Optic cable	Single mode Multi mode	100Gbps	Adv: Ultra-high speed, extreme bandwidth Disadv: Expensive material, requires specialized installations & skills	Backbone networks, long-distance data transmission

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Part-B Wiring Standards (Pinouts)

- Straight-Through Cable: Both ends use identical standards. Used to connect different device types
- Crossover Cable: One end uses Standard A and the other uses Standard B. Used to connect similar device types (eg., PC to PC)

Pin#	Wire Color Code (Std A)	Wire Color Code (Std B)	Signal Function
1	White/Green	White/Orange	Transmit+ (TX+)
2	Green	Orange	Transmit- (TX-)
3	White/Orange	White/Green	Receive+ (RX+)
4	Blue	Blue	Unused/Reserved
5	White/Blue	White/Blue	Unused/Reserved
6	Orange	Green	Receive (RX-)
7	White/Brown	White/Brown	Unused/Reserved
8	Brown	Brown	Unused/Reserved

Step-by-Step Cable Assembly Procedure:

* Thread Protective Boots: Slide the rubber connector boots onto both ends of the cable before stripping
* Strip Cable Outer Jacket: Use the wire-stripping section of the crimping tool to remove roughly 1.5 cm (0.6 inches) of the outer PVC sheath from the end.
* Untangle & Arrange Wires: Untwist the 4 pairs of wire and straighten them out in the designated pinout order
* Trim to Uniform length: Flatten the ordered wires side-by-side and snip them straight across so all conductor ends align evenly.

* Insert the RJ45 connector: Orient the RJ45 plug with the release clip facing down. Push the wires firmly into the plug until each conductor reaches the front edge and the outer jacket sits beneath the cable retention & notch.

* Crimp the Plug: Insert the RJ45 plug into the crimping slot of the tool and squeeze the handles firmly until the pins pierce the wires and the cable clips locks into place.

* Terminate Second End: Repeat steps 2-6 for the opposite end of the cable.

Testing & Verification:

* Plug both ends of the constructed cable into network interface cards (NICs) on two computers to test point-to-point data communication, or connect to a cable tester to verify continuous pin alignment.

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